Your grade: 85.71%

Your latest: **85.71%** • Your highest: **85.71%** • To pass you need at least 80%. We keep your highest score.

Next item

Instructions

1. What does scalar multiplication in a vector space involve? 1 / 1 point

- Multiplying two vectors to produce another vector
- Dividing two scalars to produce a vector
- Adding two vectors to produce a scalar
- Multiplying a scalar and a vector to produce another vector

Nice work

Correct! Scalar multiplication involves multiplying a scalar and a vector to produce another vector.

2. True or False: The set of all 2×2 real matrices forms a vector space under matrix addition and scalar multiplication. 1 / 1 point

- True
- False

Nice work

Correct! The set of all 2x2 real matrices forms a vector space under matrix addition and scalar multiplication.

3. In a vector space V, if v is any vector and α is a scalar, which of the following properties is described by $\alpha \cdot (u + v) = \alpha \cdot u + \alpha \cdot v$? 1 / 1 point

- Commutativity of addition
- Distributivity of scalar multiplication over vector addition
- Identity element of addition
- Associativity of addition

Nice work

Correct! The property described is the distributivity of scalar multiplication over vector addition.

4. True or False: The set of all diagonal matrices is a subspace of the vector space of all square matrices. 1 / 1 point

- True
- False

Nice work

Correct! The set of diagonal matrices satisfies all vector space properties, including closure under addition and scalar multiplication.

5. True or False: If U is a subset of a vector space V, then U is automatically a subspace. 1 / 1 point

- True
- False

Nice work

Correct! A subset must also satisfy closure under addition and scalar multiplication and include the zero vector to be a subspace.

6. Consider the vector space R^3. Which of the following sets is NOT a subspace of R^3? 1 / 1 point

- The set containing only the zero vector 0=(0,0,0)
- The set of all vectors where x=0
- The set of all vectors on the plane x+y+z=0
- The set of all vectors (x,y,z) such that x=1

Nice work

Correct! The set of all vectors ((x, y, z)) such that (x = 1) is not a subspace because it does not include the zero vector and is not closed under scalar multiplication.

7. Which of the following is NOT a property of vector addition in a vector space? 0 / 1 point

- Associativity of addition
- Distributivity of scalar multiplication over vector addition
- Identity element of addition
- Commutativity of addition

Try again

Review the properties of vector addition and try again.